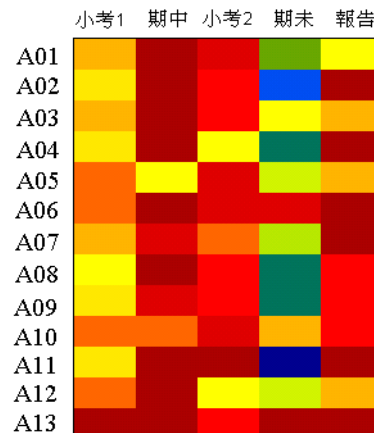
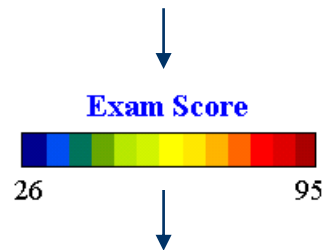


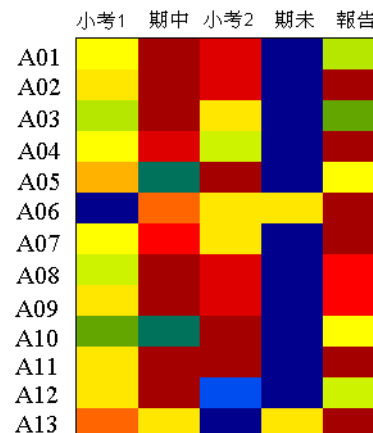
Visualization

Heat Map

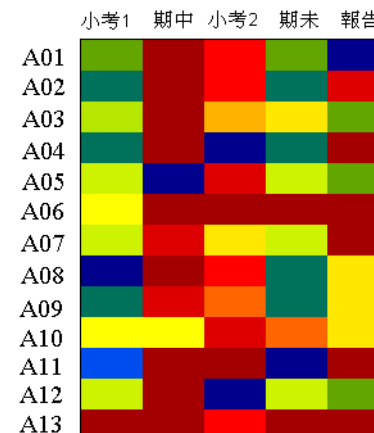
	A	B	C	D	E	F
1	學號	小考1	期中考	小考2	期末考	報告
2	A01	69	92	85	45	62
3	A02	66	90	83	36	90
4	A03	72	92	80	62	70
5	A04	68	90	60	37	95
6	A05	74	60	86	54	70
7	A06	77	90	88	88	95
8	A07	73	88	77	51	95
9	A08	61	90	84	40	82
10	A09	66	88	82	39	80
11	A10	76	75	87	72	80
12	A11	64	90	90	26	95
13	A12	75	90	60	55	70
14	A13	92	90	83	90	95



Matrix condition



Row Condition

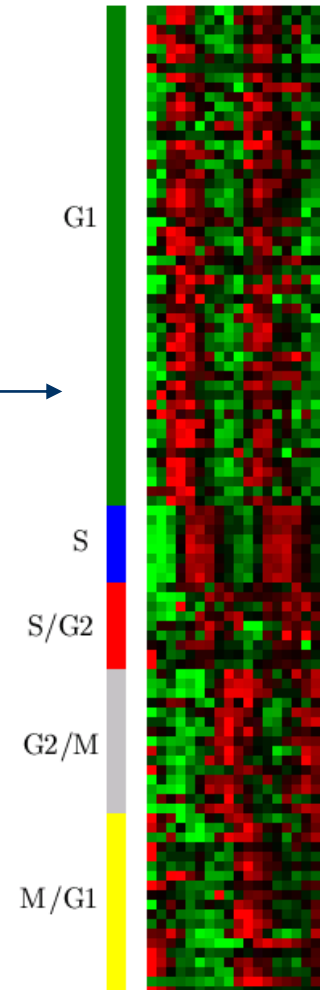
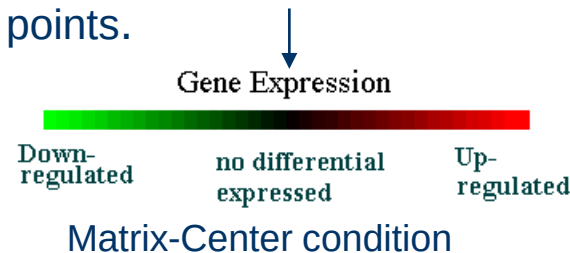


Column Condition

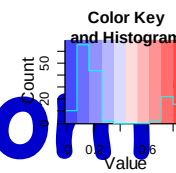
Microarray Data of Yeast Cell Cycle

■ Synchronized by alpha factor arrest method (Spellman et al. 1998; Chu et al. 1998)

■ 103 known genes: every 7 minutes and totally 18 time points.

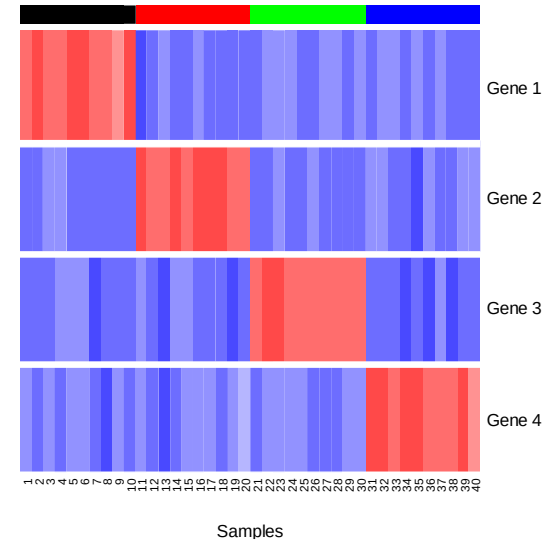


Heat Map (cont'd)



```
set.seed(1024)
my.samples<-c(rep(1,10),rep(2,10),rep(3,10),rep(4,10))
ng<-max(my.samples)
ns<-length(my.samples)
my.data<-NULL
for(i in 1:ng){
  my.exprs<-rnorm(ns,0.15,0.05)
  my.exprs[which(my.samples==i)]<-rnorm(length(which(my.samples==i)),0.8,0.05)
  my.data<-rbind(my.data,my.exprs)
}
rownames(my.data)<-paste("Gene",1:4)

library(gplots)
my.colors<-as.character(factor(my.samples,label=c("black","red","green","blue")))
X11()
heatmap.2(my.data, Rowv=F, Colv=F, dendrogram="none", col=bluered(16)[-c(1:2,8:9,15:16)],
  xlab="Samples", ColSideColors=my.colors, trace="none", cexRow=1.5, cexCol=1,
  rowsep=1:4, sepwidth=c(0.05,0.05))
```



S1 S2 S3 Sk

Sub-type 1

$$\begin{cases} G11=(0.7, 0.8, 0.9, \dots, 0.1) \\ G12=(0.8, 0.9, 0.7, \dots, 0.2) \\ G13=(0.9, 0.7, 0.7, \dots, 0.3) \end{cases}$$

$$\begin{cases} G11=(0.7, 0.8, 0.9, \dots, 0.1) \\ G21=(0.1, 0.2, 0.1, \dots, 0.1) \end{cases}$$

Sub-type 2

$$\begin{cases} G21=(0.1, 0.2, 0.1, \dots, 0.1) \\ G22=(0.2, 0.3, 0.2, \dots, 0.2) \\ G23=(0.1, 0.2, 0.3, \dots, 0.3) \end{cases}$$

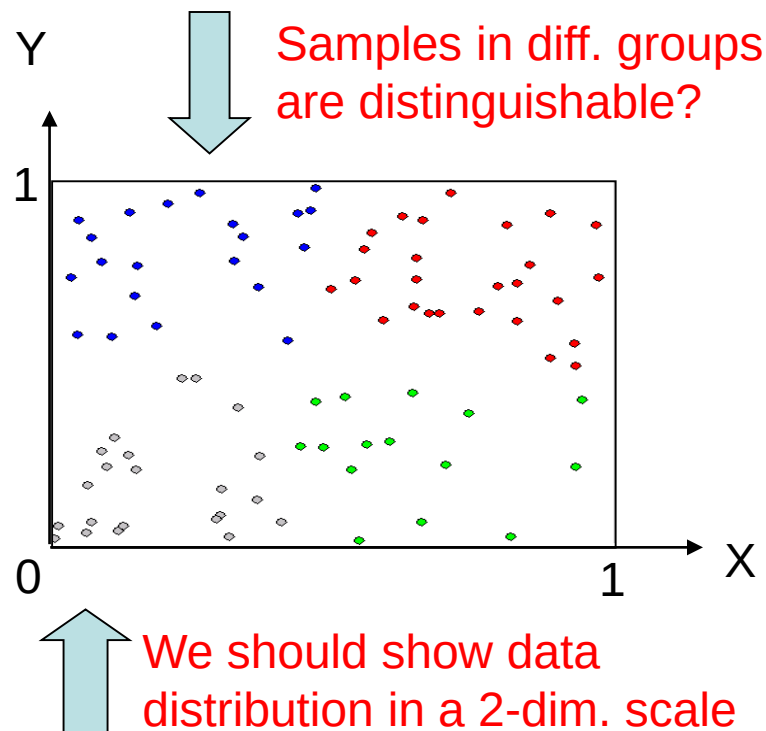
Sub-type 3

$$\begin{cases} G31=(0.3, 0.1, 0.2, \dots, 0.1) \\ G32=(0.2, 0.2, 0.1, \dots, 0.2) \\ G33=(0.4, 0.3, 0.3, \dots, 0.3) \end{cases}$$

Sub-type 4

$$\begin{cases} G41=(0.3, 0.1, 0.2, \dots, 0.9) \\ G42=(0.1, 0.3, 0.3, \dots, 0.8) \\ G43=(0.2, 0.1, 0.1, \dots, 0.9) \end{cases}$$

...



$$\begin{cases} G11=(0.7, 0.8, 0.9, \dots, 0.1) \\ G12=(0.8, 0.9, 0.7, \dots, 0.2) \\ G21=(0.1, 0.2, 0.1, \dots, 0.1) \\ G31=(0.3, 0.1, 0.2, \dots, 0.1) \\ G41=(0.3, 0.1, 0.2, \dots, 0.9) \end{cases}$$

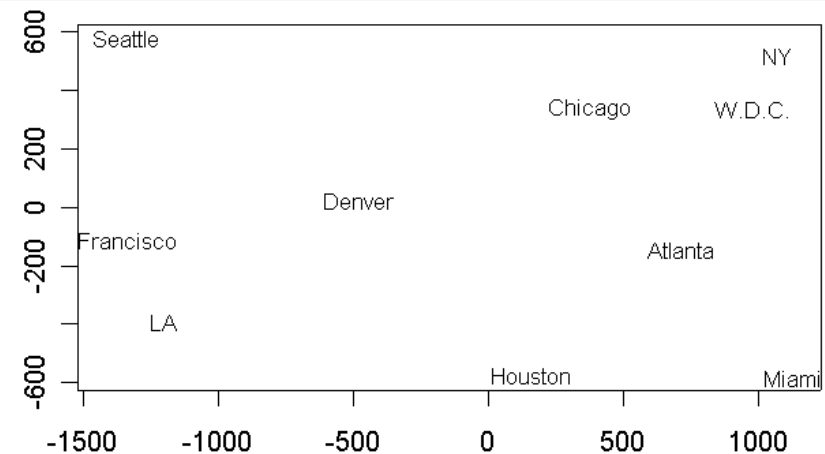
Multidimensional Scaling (MDS)

(Torgerson 1952; Cox and Cox 2001)

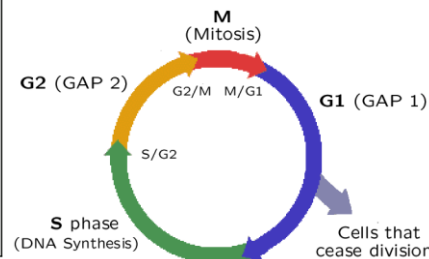
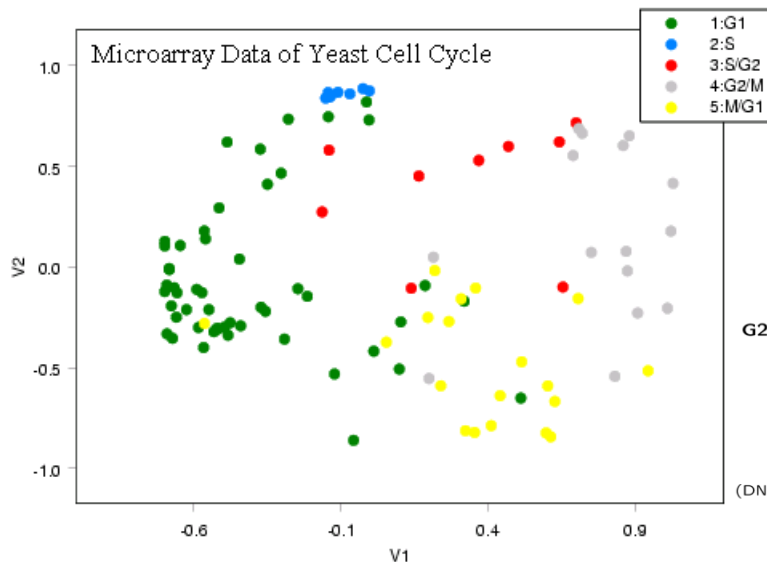
Multidimensional scaling takes a set of dissimilarities and returns a set of points such that the distances between the points are approximately equal to the dissimilarities.

Analysis of Flying Mileages Between Ten U.S. Cities

0										Atlanta
587	0									Chicago
1212	920	0								Denver
701	940	879	0							Houston
1936	1745	831	1374	0						Los Angeles
604	1188	1726	968	2339	0					Miami
748	713	1631	1420	2451	1092	0				New York
2139	1858	949	1645	347	2594	2571	0			San Francisco
2182	1737	1021	1891	959	2734	2408	678	0		Seattle
543	597	1494	1220	2300	923	205	2442	2329	0	Washington D.C.



2D MDS configuration plot for 103 known genes

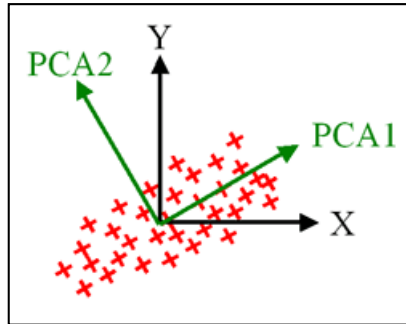


Microarray Data of Yeast Cell Cycle
Synchronized by alpha factor arrest method
(Spellman et al. 1998; Chu et al. 1998)

103 known genes: every 7 minutes and
totally 18 time points.

Principal Component Analysis (PCA)

(Pearson 1901; Hotelling 1933; Jolliffe 2002)



The i th principal component of $\mathbf{X}$ is $\mathbf{X}'\mathbf{v}_i$, where $\mathbf{v}_i$ is the i th normalized eigenvector of $\Sigma_{\mathbf{X}}$ corresponding to the i th largest eigenvalue.

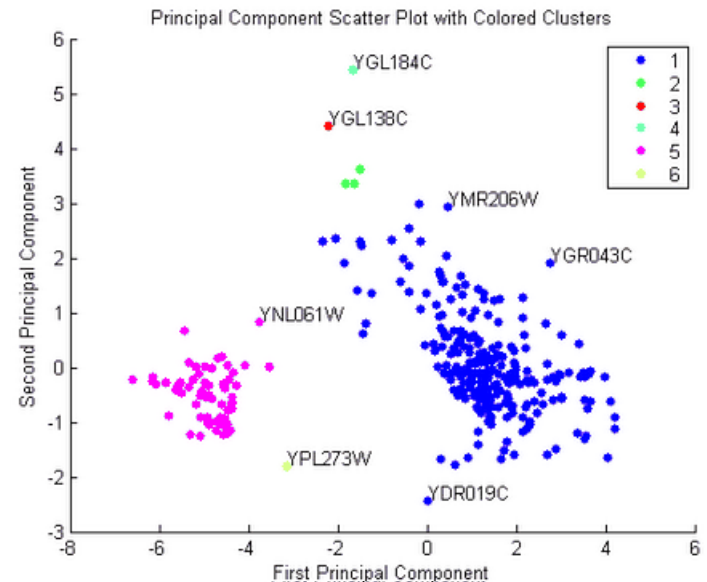
The PCA summarizes the dispersion of data points as data cloud in a small number of major axes (principal components) of variation among the variables.

Goal: to reduce the dimensionality of the data matrix by finding the new variables.

Cumulative Sum of the Variances:

1	78.3719
2	89.2140
3	93.4357
4	96.0831
5	98.3283
6	99.3203
7	100.0000

This shows that almost 90% of the variance is accounted for by the first two principal components.



Yeast Microarray Data is from

DeRisi, JL, Iyer, VR, and Brown, PO.(1997). "Exploring the metabolic and genetic control of gene expression on a genomic scale"; Science, Oct 24;278(5338):680-6.

Missing value imputation (KNN) & PCA

```
## Yeast Cell Cycle Data, column 1: gene name, column 2~78: data, column 79: peak  
> data<-read.table("cycleData_peak.txt", sep="\t", header=T, as.is=T)
```

```
##### Missing value imputation (KNN)
```

```
> BiocManager::install("impute")  
> library(impute)  
> data[,2:78]<- impute.knn(as.matrix(data[, 2:78]))$data
```

```
##### PCA, project samples
```

```
> data.prcomp1<-prcomp(t(data[,2:78]), scale=T, retx=T)  
> summary(data.prcomp1)  
> str(data.prcomp1)
```

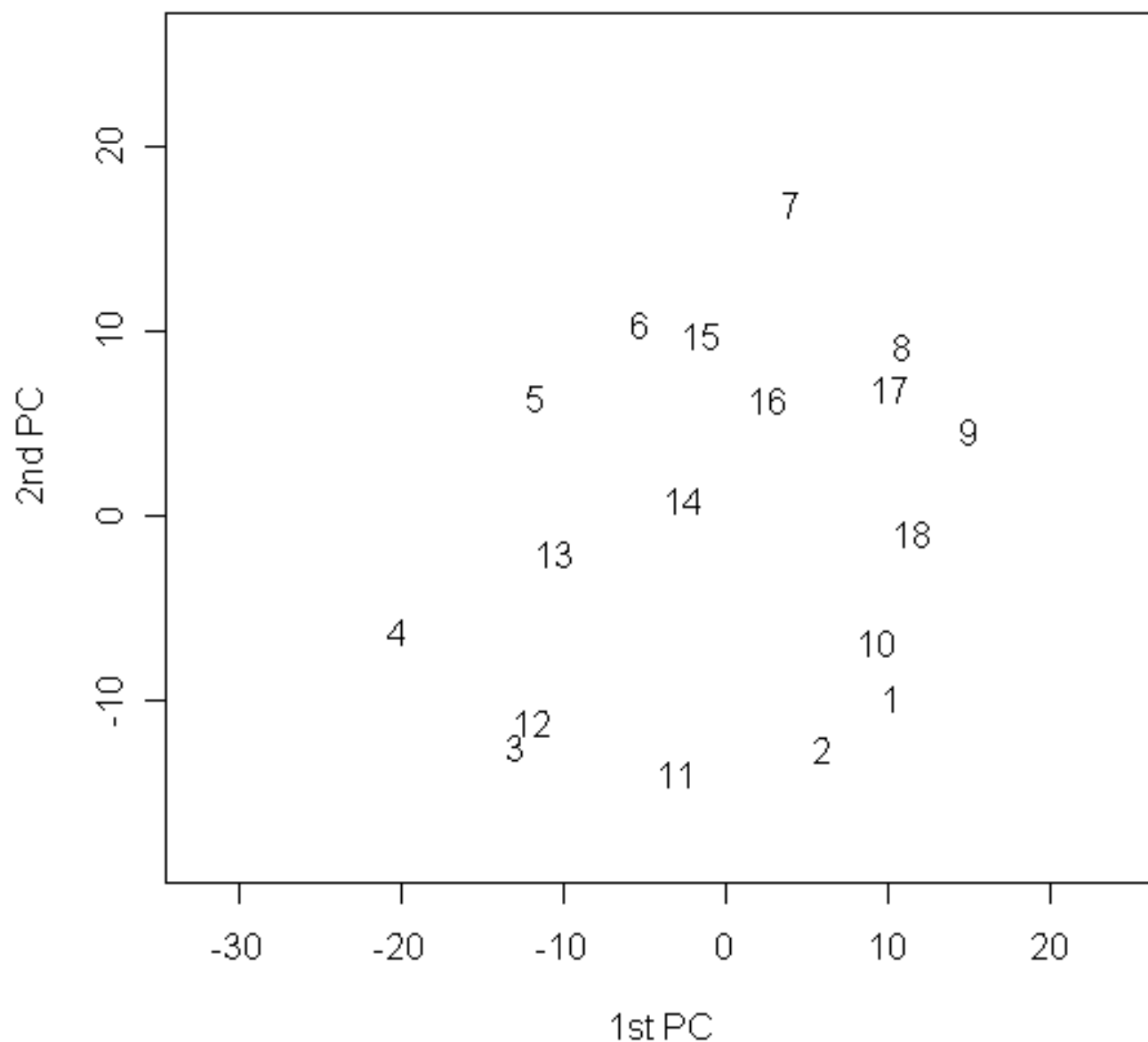
```
## alpha ##
```

```
> plot(c(min(data.prcomp1$x[,1]), max(data.prcomp1$x[,1])), c(min(data.prcomp1$x[,2]),  
  max(data.prcomp1$x[,2])), type="n", xlab="1st PC", ylab="2nd PC")  
> text(data.prcomp1$x[5:22,1], data.prcomp1$x[5:22,2], 1:18)
```

```
## cdc ##
```

```
> plot(c(min(data.prcomp1$x[,1]), max(data.prcomp1$x[,1])), c(min(data.prcomp1$x[,2]),  
  max(data.prcomp1$x[,2])), type="n", xlab="1st PC", ylab="2nd PC")  
> text(data.prcomp1$x[23:63,1], data.prcomp1$x[23:63,2], 1:41)
```

Missing value imputation (KNN) & PCA (cont'd)



Missing value imputation (KNN) & PCA (cont'd)

```
#### PCA, project genes
```

```
> data.prcomp2<-prcomp(data[,2:78], scale=T, retx=T)
```

```
> summary(data.prcomp2)
```

```
> str(data.prcomp2)
```

```
> plot(c(min(data.prcomp2$x[,1]), max(data.prcomp2$x[,1])), c(min(data.prcomp2$x[,2]),  
  max(data.prcomp2$x[,2])), type="n", xlab="1st PC", ylab="2nd PC")
```

```
> text(data.prcomp2$x[data[,79]=="G1",1], data.prcomp2$x[data[,79]=="G1",2], 1, col=1)
```

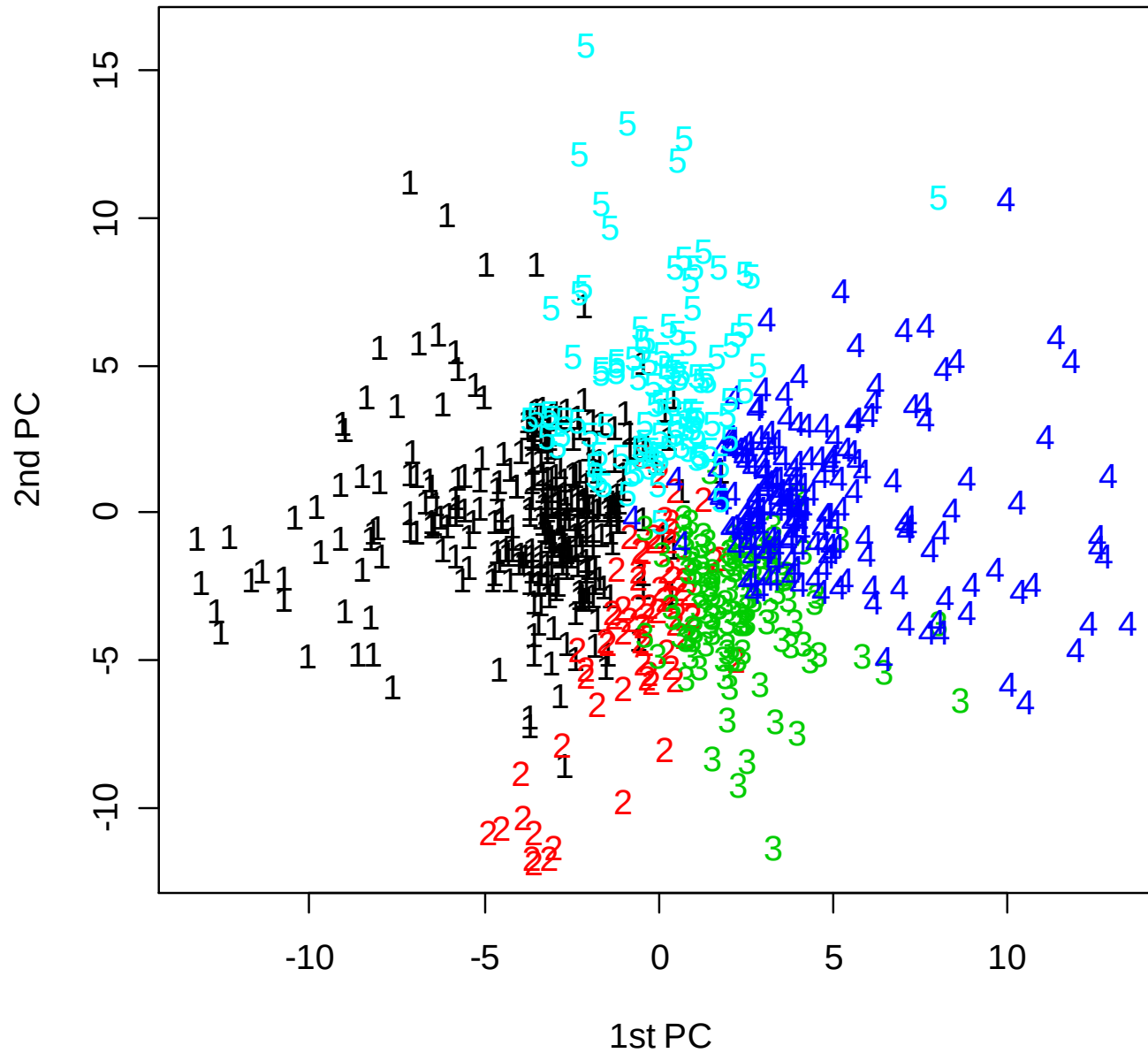
```
> text(data.prcomp2$x[data[,79]=="S",1], data.prcomp2$x[data[,79]=="S",2], 2, col=2)
```

```
> text(data.prcomp2$x[data[,79]=="S/G2",1], data.prcomp2$x[data[,79]=="S/G2",2], 3, col=3)
```

```
> text(data.prcomp2$x[data[,79]=="G2/M",1], data.prcomp2$x[data[,79]=="G2/M",2], 4, col=4)
```

```
> text(data.prcomp2$x[data[,79]=="M/G1",1], data.prcomp2$x[data[,79]=="M/G1",2], 5, col=5)
```

Missing value imputation (KNN) & PCA (cont'd)



Multidimensional Scaling (MDS)

1. project samples

```
> data.mds1<-cmdscale(dist(t(data[,2:78])), 2)
```

```
> row.names(data.mds1)
```

```
## alpha ##
```

```
> plot(c(min(data.mds1[,1]), max(data.mds1[,1])), c(min(data.mds1[,2]),  
  max(data.mds1[,2])), type="n", xlab="1st direction", ylab="2nd direction")
```

```
> text(data.mds1[5:22,1], data.mds1[5:22,2], 1:18)
```

```
## cdc ##
```

```
> plot(c(min(data.mds1[,1]), max(data.mds1[,1])), c(min(data.mds1[,2]),  
  max(data.mds1[,2])), type="n", xlab="1st direction", ylab="2nd direction")
```

```
> text(data.mds1[23:63,1], data.mds1[23:63,2], 1:41)
```

2. project genes

```
> data.mds2<-cmdscale(dist(data[,2:78]), 2)
```

```
> plot(c(min(data.mds2[,1]), max(data.mds2[,1])), c(min(data.mds2[,2]),  
  max(data.mds2[,2])), type="n", xlab="1st direction", ylab="2nd direction")
```

```
> text(data.mds2[data[,79]=="G1",1], data.mds2[data[,79]=="G1",2], 1, col=1)
```

```
> text(data.mds2[data[,79]=="S",1], data.mds2[data[,79]=="S",2], 2, col=2)
```

```
> text(data.mds2[data[,79]=="S/G2",1], data.mds2[data[,79]=="S/G2",2], 3, col=3)
```

```
> text(data.mds2[data[,79]=="G2/M",1], data.mds2[data[,79]=="G2/M",2], 4, col=4)
```

```
> text(data.mds2[data[,79]=="M/G1",1], data.mds2[data[,79]=="M/G1",2], 5, col=5)
```

Clustering

Clustering vs. Classification

Two main types of microarray experiments:

- Samples are not categorized, but gene expression levels are followed with time
 - to find genes whose expression levels vary together
 - Clustering
- Samples are labelled as either normal or diseased tissues
 - to find genes whose expression levels can distinguish different labels
 - Classification

Clustering Analysis (Unsupervised Learning)

Clustering analysis is the art of finding groups in data.
(Kaufman & Rousseeuw, 1990)

■ Objective of clustering analysis :

one wants to form groups in such that

1. **homogeneous**

objects in the same group are similar to each other

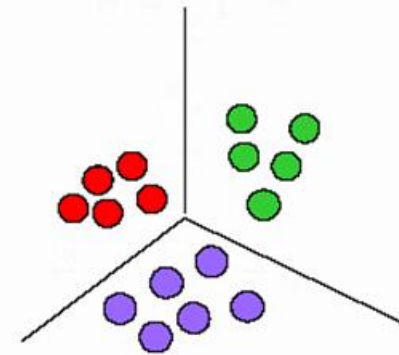
2. **well separated**

objects in different groups are dissimilar as possible

Clustering Analysis (co

Partitioning

- Clustering is the representation of distance measurements between objects.
- The main goal of clustering is to use **similarity** or **distance measurements** between objects to represent them.
- Data points within a cluster are more similar, and those in separate cluster are less similar.

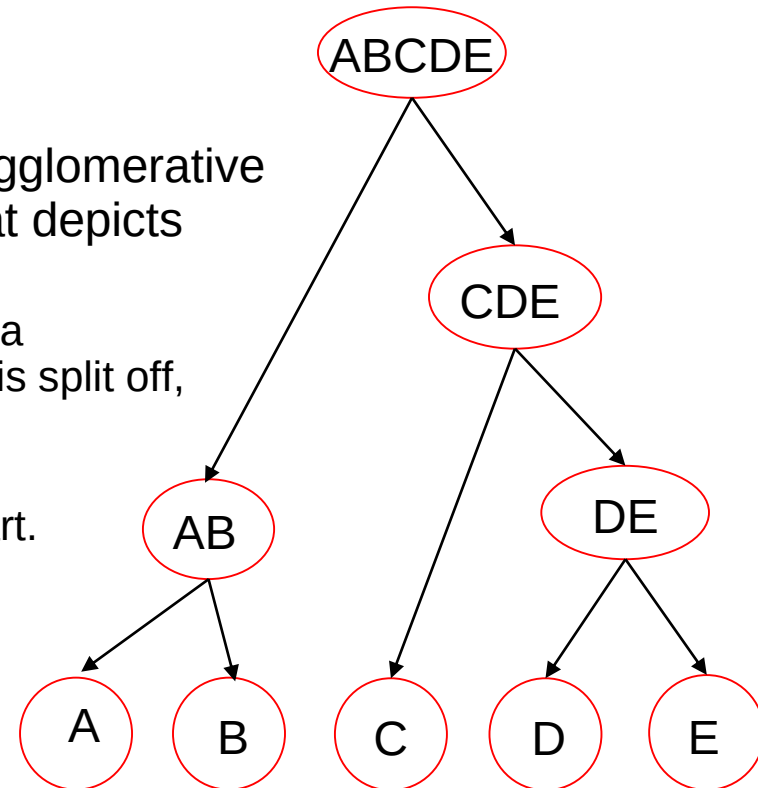


- **Hierarchical clustering** can be performed using agglomerative and divisive approaches. The result is a tree that depicts the relationships between the objects.

➤ **Divisive clustering:** begin at step 1 with all the data in one cluster, in each subsequent step a cluster is split off, until there are n clusters.

➤ **Agglomerative clustering:** all the objects start apart. There are n clusters at step 0, each object forms a separate cluster. In each subsequent step two clusters are merged, until only one cluster is left.

- **Non-Hierarchical clustering**



Hierarchical Clustering and Dendrogram (Kaufman and Rousseeuw, 1990)

Example:

Average linkage clustering

	a	b	c	d	e
a	0	2	6	10	9
b		0	5	9	8
c			0	4	5
d				0	3
e					0

	{a, b}	c	d	e
{a, b}	0	5.5	9.5	8.5
c		0	4	5
d			0	3
e				0

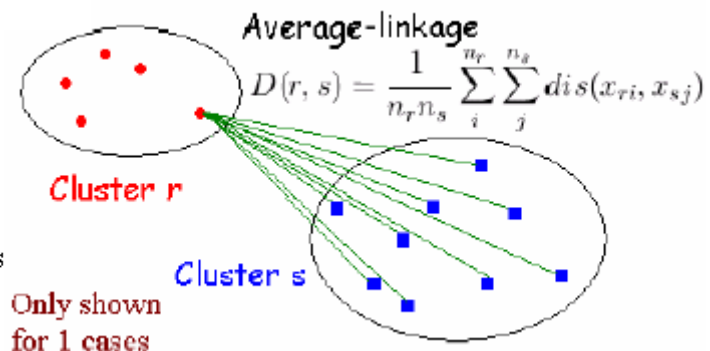
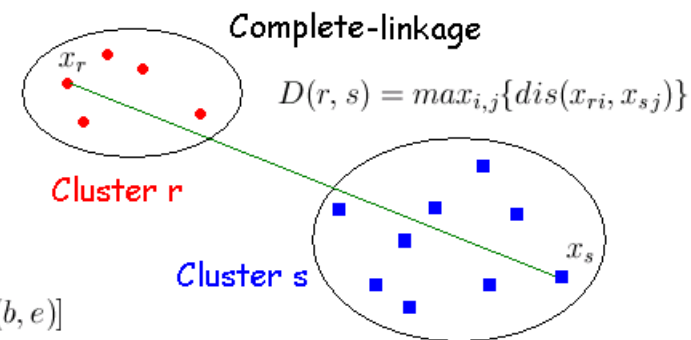
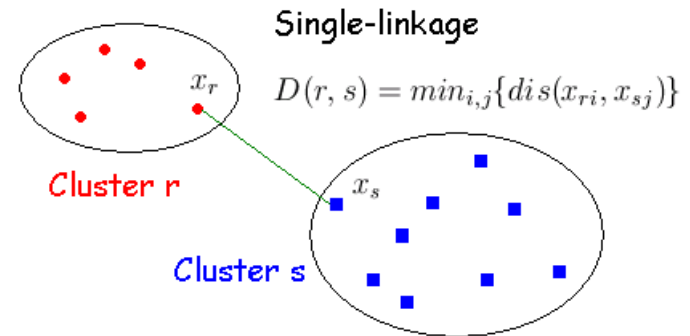
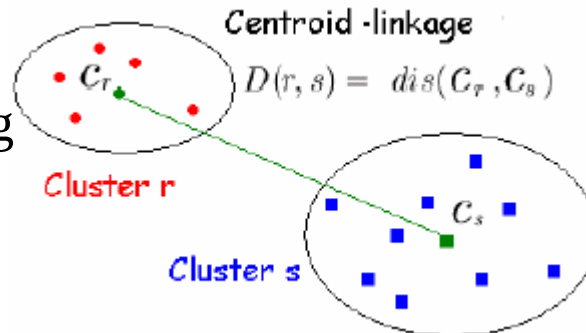
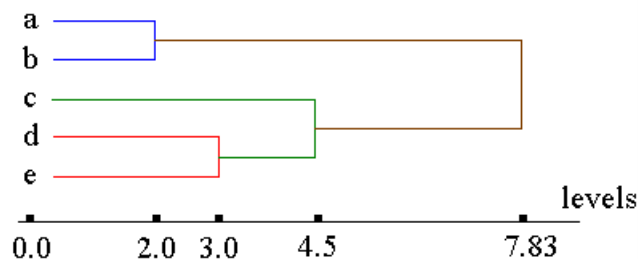
	{a, b}	c	{d, e}
{a, b}	0	5.5	9.0
c		0	4.5
{d, e}			0

	{a, b}	{c, d, e}
{a, b}	0	7.83
{c, d, e}		0

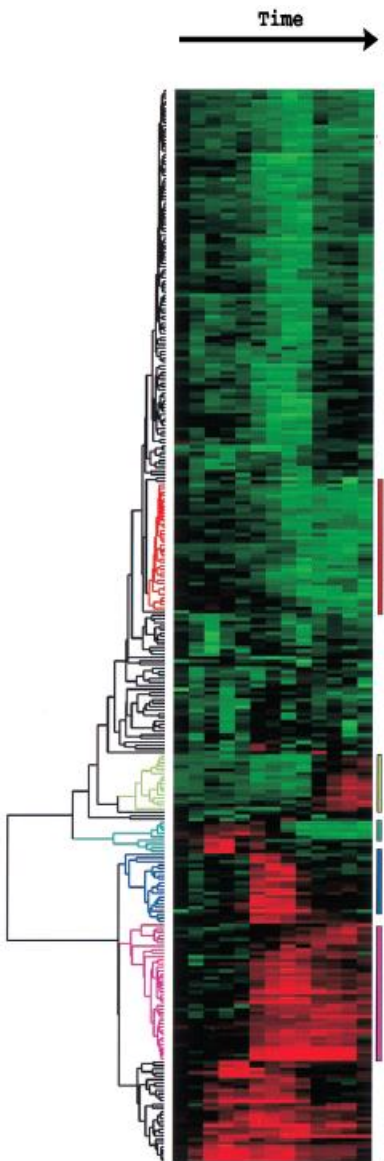
$$D(\{a, b\}, \{c\}) = \frac{1}{2}[D(a, c) + D(b, c)] = \frac{1}{2}(6 + 5) = 5.5$$

$$D(\{a, b\}, \{d, e\}) = \frac{1}{4}[D(a, d) + D(a, e) + D(b, d) + D(b, e)]$$

$$= \frac{1}{4}(10 + 9 + 9 + 8) = 9$$



Hierarchical Clustering



Proc. Natl. Acad. Sci. USA
Vol. 95, pp. 14863–14868, December 1998
Genetics

Cluster analysis and display of genome-wide expression patterns

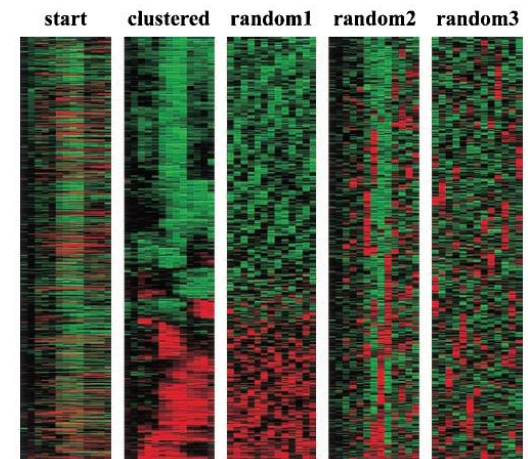
MICHAEL B. EISEN*, PAUL T. SPELLMAN*, PATRICK O. BROWN†, AND DAVID BOTSTEIN*‡

FIG. 1. Clustered display of data from time course of serum stimulation of primary human fibroblasts. Experimental details are described elsewhere (11). Briefly, foreskin fibroblasts were grown in culture and were deprived of serum for 48 hr. Serum was added back and samples taken at time 0, 15 min, 30 min, 1 hr, 2 hr, 3 hr, 4 hr, 8 hr, 12 hr, 16 hr, 20 hr, 24 hr. The final datapoint was from a separate unsynchronized sample. Data were measured by using a cDNA microarray with elements representing approximately 8,600 distinct

human genes. All measurements are relative to time 0. Genes were selected for this analysis if their expression level deviated from time 0 by at least a factor of 3.0 in at least 2 time points. The dendrogram and colored image were produced as described in the text; the color scale ranges from saturated green for log ratios -3.0 and below to saturated red for log ratios 3.0 and above. Each gene is represented by a single row of colored boxes; each time point is represented by a single column. Five separate clusters are indicated by colored bars and by identical coloring of the corresponding region of the dendrogram. As described in detail in ref. 11, the sequence-verified named genes in these clusters contain multiple genes involved in (A) cholesterol biosynthesis, (B) the cell cycle, (C) the immediate-early response, (D) signaling and angiogenesis, and (E) wound healing and tissue remodeling. These clusters also contain named genes not involved in these processes and numerous uncharacterized genes. A larger version of this image, with gene names, is available at <http://rana.stanford.edu/clustering/serum.html>.

Software: Cluster and TreeView

FIG. 3. To demonstrate the biological origins of patterns seen in Figs. 1 and 2, data from Fig. 1 were clustered by using methods described here before and after random permutation within rows (random 1), within columns (random 2), and both (random 3).



K-means Clustering

- K-means is a partition methods for clustering.
- Data are classified into k groups as specified by the user.
- Two different clusters cannot have any objects in common, and the k groups together constitute the full data set.

Optimization problem:

Minimize the sum of squared within-cluster distances

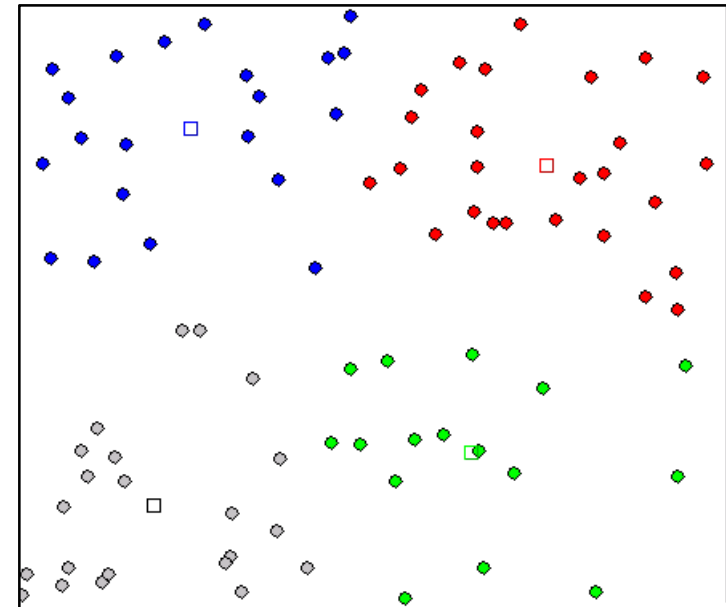
The K-Means Algorithm

1. The data points are randomly assigned to one of the K clusters.
2. The position of the K centroids are determined (initial group centroids).
3. For each data point:
 - Calculate the distance from the data point to each cluster.
 - Assign data point to the cluster that has the closest centroid.
4. Repeat the above step until the centroids no longer move.

The choice of initial partition can greatly affect the final clusters that result.

$$W(C) = \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=C(j)=k} d_E(x_i, x_j)^2$$

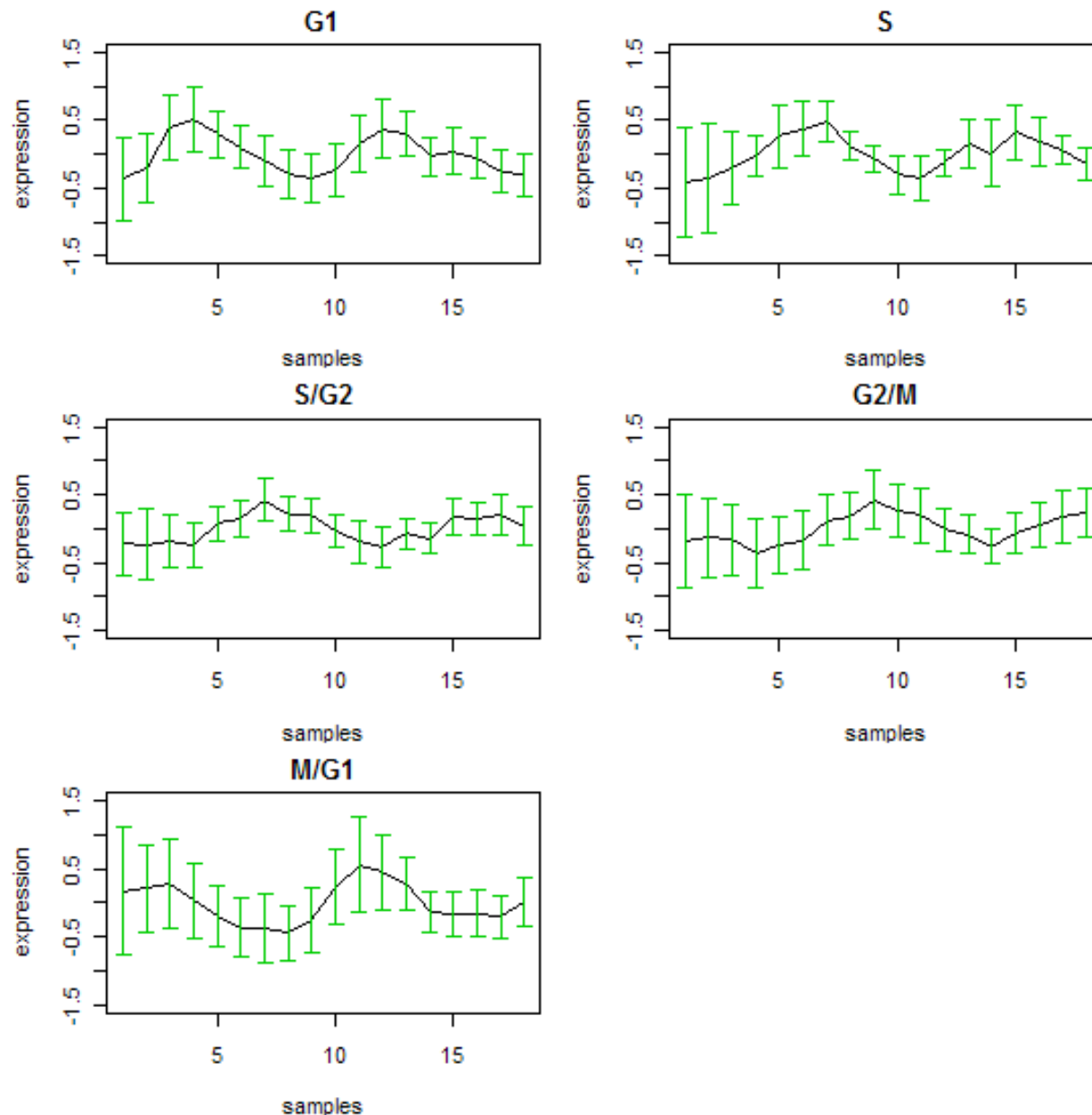
Converged



Clustering Analysis

```
##### Plot cluster pattern by functional annotation
fun.annot<-c("G1", "S", "S/G2", "G2/M", "M/G1")
par(mfrow=c(3,2), mar=c(4,4,2,1))
for(i in 1:5)
{
  subdata<-data[data[,79]==fun.annot[i], 6:23]
  plot(c(1, ncol(subdata)), c(-1.5,1.5), type="n", xlab="samples", ylab="expression")
  col.std<- apply(subdata, 2, sd)
  col.mean<- apply(subdata, 2, mean)
  lines(1:ncol(subdata), col.mean)
  for(j in 1:ncol(subdata))
  {
    lines(c(j, j), c(col.mean[j]-col.std[j], col.mean[j]+col.std[j]), col=3)
    lines(c(j-0.3, j+0.3), c(col.mean[j]-col.std[j], col.mean[j]-col.std[j]), col=3)
    lines(c(j-0.3, j+0.3), c(col.mean[j]+col.std[j], col.mean[j]+col.std[j]), col=3)
  }
  title(fun.annot[i])
}
```

Clustering Analysis (cont'd)



Clustering Analysis (cont'd)

Clustering: hierarchical clustering

```
par(mfrow=c(2,1))
```

```
data.hclust<-hclust(dist(t(data[, 6:23])) , method= "complete")
```

```
plot(data.hclust)
```

```
data.hclust<-hclust(dist(t(data[, 6:23])), method="single")
```

```
plot(data.hclust)
```

Clustering: kmeans

```
data.kmeans<-kmeans(as.matrix(data[,2:78]), 10) ## k=10
```

```
table(as.factor(data[,79]), as.factor(data.kmeans$cluster))
```

```
clustered.data<-as.matrix(data[data.kmeans$cluster==1,2:78])
```

```
for(i in 2:10)
```

```
{
```

```
  clustered.data<-rbind(clustered.data, matrix(NA, 3, ncol(clustered.data)))
```

```
  clustered.data<-rbind(clustered.data, as.matrix(data[data.kmeans$cluster==i, 2:78]))
```

```
}
```

```
heatmap(as.matrix(clustered.data) , Rowv=NA, Colv=NA, scale= "none",
```

```
  col=rgb(c(rep(0, 16), 0, 1:16)/16, c(16:1, 0, rep(0, 16))/16, rep(0, 33)))
```

http://geography.uoregon.edu/datagraphics/color_scales.htm

HW5

- Please obey the following rules and finish the HW:
 - **Perform a 2D MDS & PCA plots**
 - Please use 30 DE genes (in HW4)
 - Please use three different colors to represent three types of samples
 - **Perform a plot including heatmap and dendrogram**
 - Please use 30 DE genes (in HW4)
 - Conditions: Average linkage and Euclidean distance

Hint:

```
my.dist<-function(x){  
  dist(x,method="euclidean")}
```

```
my.hclust<-function(d){  
  hclust(d,method="average")}
```

```
heatmap.2(..., distfun=my.dist, hclustfun=my.hclust,...)
```